



PROFESSIONAL CERTIFICATE IN DATA SCIENCE AND ANALYTICS

Advance your analytics career with cutting-edge tools and techniques

Delivered in collaboration with  EMERITUS

Overview

Data gives organizations an edge, enabling robust decision-making based on the most recent information. Every day, more organizations are adopting data-driven mindsets. Today, a staggering 97% of business executives report their organizations are investing in big data and AI projects. So what changed?

Recognizing the power of being data-informed and the threat of technology-driven competitors, executives are turning to data to guide their decisions and to data professionals to deliver actionable insights for decision-making.

Analytics professionals play a pivotal role in modern-day operations. They are trend-spotters, problem-solvers, communication specialists, and advisors. They are a unique breed – and a valued organizational asset. So, it is no surprise that the outlook for data professionals is bright. By 2031, the percentage of data scientists and business analysts employed in the U.S. is projected to increase by 36%, according to the U.S. Bureau of Labor Statistics.

Developed by expert faculty from MIT, MIT xPRO's Professional Certificate in Data Science and Analytics is designed to meet the growing demand for data professionals who have the technical skills to transform data into insights that can facilitate decision-making. The program imparts a strong foundation in data science and analytics, giving you the skills you need to apply models, analyze data, communicate results, and ultimately help your organization identify optimal solutions to modern-day business challenges.

USD 120,000

Average salary for data scientists
in the United States in 2022

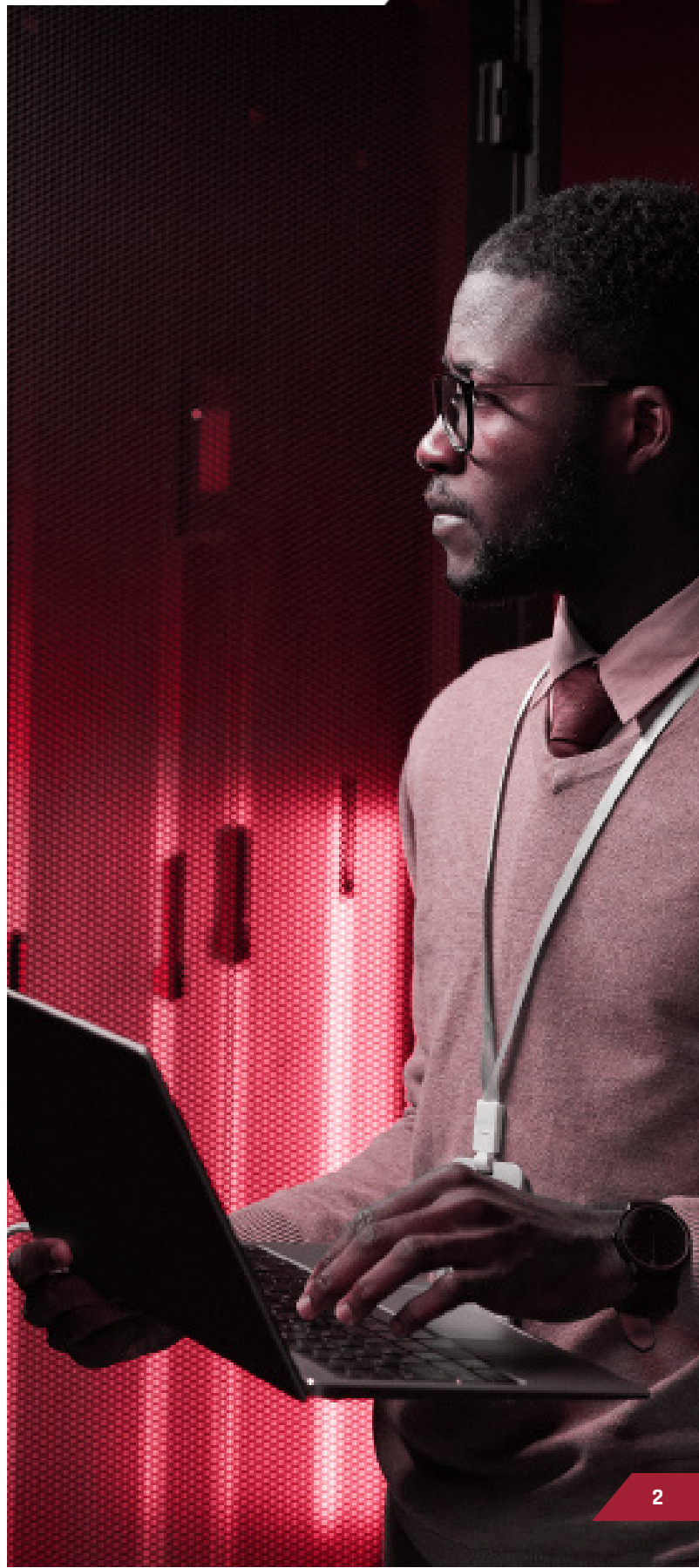
(Source: Glassdoor)

PRICE USD 7,450

DURATION

6 months, Online

15-20 hours per week



Program Highlights



Earn an MIT xPRO certificate and 36 continuing education units (CEUs)



Gain insights from MIT faculty, experts in the field who bring industry experience to the classroom



Learn from real-world case studies solved by faculty using applications of data science and applied analytics



Demonstrate your knowledge in a final capstone project by developing a portfolio of application-based assignments completed throughout the program



Receive one-on-one career support from Emeritus and introductions to hiring partners for eligible participants



Program Experience



20+ hours
of prerecorded
MIT faculty videos



5+ hours
of live mentorship
and career support



8 hours
of optional career
development
activities



19 career
development
video lectures
covering **30**
career topics



Access to résumé
referrals to Emeritus
employer partners
for up to **6 months**
post-program

Sample Weekly Program Planner

Learners should expect to dedicate a minimum of **15-20 hours** per week to the program.



1 hour of recorded video lectures
with faculty



5-8 hours of rigorous, graded assignments
to apply and reinforce lecture material



2 hours of interaction with program
learning facilitators



2-3 hours of engaging group discussion
with peers to exchange and generate ideas



2 hours of self-study
and practice exercises

*Career Services are provided by Emeritus, a learning partner for this program.

Learning Journey

From navigating the enrollment process to identifying job opportunities, we partner with you to take the next step in your career.

LEARNING FACILITATOR

Your learning facilitator will leverage their industry experience and expertise to guide you by holding live sessions, providing assignment feedback, and answering questions.

STUDENT SUCCESS COACH

Your student success coach will support you by offering one-on-one guidance throughout the program, partnering with you to ensure you have a smooth, successful learning experience.

LEARNING COMMUNITY

Your learning community will provide an interactive environment where you can learn with a group of like-minded individuals and build a global network of peers.



PROGRAM ADVISOR

Your program advisor will be your enrollment resource, answering any pre-program questions and easing your transition into the program.

CAREER COACH

Your career coach will help you successfully navigate your job search by assisting with goal setting, providing feedback on your cover letter, résumé, and LinkedIn profile, and conducting mock interviews. They will be a source of up-to-the-minute information on hiring trends and help celebrate the next step in your career.

Who Is This Program For?

Whether you are a recent STEM graduate, an experienced data or business professional, or an entrepreneur, this program offers a singular opportunity to deepen your knowledge of data science and analytics with faculty from the #1 technical school in the U.S. By developing the skills you need to put theory into practice, you will become proficient in the science of data-based decision-making and prepare for opportunities in a rapidly growing field.

The program will be especially valuable for:



Data professionals in engineering, finance, insurance, IT, or operations with coding experience who wish to develop a suite of advanced data analysis skills they can apply in their day-to-day work to make sound business decisions that are informed by data.



Business professionals in sales, marketing, IT, or operations looking to sharpen their decision-making ability by learning to model, execute, and analyze business data and, in turn, use their data science expertise to advance their careers.



Recent graduates with a STEM background who wish to build knowledge and skills in data science to prepare for a role as an analyst and start down a career path in the field.

Prerequisites

Participants should be familiar with Excel datasets and data visualization. Experience setting up the Jupyter Notebook and basic knowledge of Python is also recommended.

Prepare for These Future Job Titles:

- Applications architect
- Business intelligence developer
- Data architect
- Data engineer
- Data Analyst
- Machine learning engineer
- Machine learning specialist

Program Curriculum

This program is designed to provide firsthand experience with the strategic applications of data to modern-day business decisions. Using tools such as Python and Google Colab, you will learn the essentials of clustering and probability, linear and logistic regression, deep learning, and digital transformation in order to optimize the decision-making process for your organization. This program's high-level key takeaways include:

- Leverage data in order to optimize or improve decision-making within an organization
- Using Python and Google Colab to train, organize, run, and analyze datasets and models that yield meaningful results
- Analyzing and translating technical results into actionable insights for executives
- Develop a portfolio of modeling tools from simple regression all the way to deep learning, neural networks, and natural language processing that showcases the hands-on application of the skills required to use data for prediction, classification, and optimization in practical business contexts.

Orientation

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Part 1

Fundamentals of Data Science

Key Takeaways

- Survey the essentials of data science including data, models, processes, and interactions with expert operators.
- Understand how data analytics and machine learning can improve decision-making, systems, and products.
- Explore how data science can be used to help businesses learn more about their customers.
- Learn how to apply, interpret, and explain key analytical frameworks.

Module 1: Introduction to Data Science

Module 2: Thinking About Risk and Uncertainty Through Probability and Distributions

Module 3: Correlation

Module 4: Clustering

Module 5: Linear Regression Part 1

Module 6: Linear Regression Part 2

Module 7: Logistic Regression

Part 2

Foundations of Optimization

Key Takeaways

- Learn about collaborative filtering models and their design.
- Discover the role optimization plays across different industries.
- Learn how to construct a linear optimization model and interpret the results.
- Understand the importance of model valuation and how it is assessed.
- Explore the steps involved in creating a model.
- Understand the value of optimization and how it can be used for the good of humanity.

Module 8: Collaborative Filtering

Module 11: Optimization Part 3

Module 9: Optimization Part 1

Module 12: Optimization Part 4

Module 10: Optimization Part 2

Module 13: Optimization Part 5

Part 3

Foundations of Machine Learning

Key Takeaways

- Learn to make predications with classification and regression trees.
- Practice performing a CART analysis and interpreting the results.
- Explore ensemble learning and how it can be used to improve predictive performance.

Module 14: Nonlinear Models for Regression and Classification: Classification and Regression Trees

Module 16: Fairness and Bias Issues in Data-Driven Predictions

Module 15: Nonlinear Models for Regression and Classification: Ensemble Learning

Part 4

Advanced Machine Learning

Key Takeaways

- Explore deep learning, and examine developments in the field.
- Learn about neural network applications, training deep neural networks, and using Keras and Tensorflow to code predictive applications.
- Discover how to apply transfer learning to represent images that a computer can understand.
- Examine the mechanics and capabilities of natural language processing.
- Learn how to use a transformer to increase the accuracy of a dataset.

Module 17: Neural Networks Part 1

Module 18: Neural Networks Part 2

Module 19: Neural Networks Part 3

Module 20: Natural Language Processing (NLP) Part 1

Module 21: Natural Language Processing (NLP) Part 2

Module 22: Interpretability and Causality in Models

Part 5

Deployment

Key Takeaways

- Discover real-world applications of AI/ML.
- Explore new applications of digital transformation.
- Develop a portfolio of assignments for a final capstone project.

Module 23: Data, Models, and Decisions

Module 24: Leading Digital Transformations

Case Studies and Examples

Throughout the program, each faculty member will present case studies they have personally helped solve using the principles and tools of data science, including:

Retail Management

Analyze how a Boston, MA retailer handled their response to the COVID-19 pandemic using the systems approach.



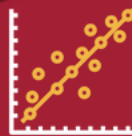
Filatoi Riuniti

Solve an optimization problem by developing a model and making recommendations on how this Italian yarn manufacturer should outsource production in order to maximize profits.



Facial Analysis Algorithms

Examine the ramifications of poorly designed models, and discover how to detect, diagnose, and mitigate biases that can arise in model-based, data-driven decision-making.



BlueBike

See how BlueBike used regression to predict demand. Then, build your own models with regression, and use R^2 to pick the optimal model.

Assignments and Activities



Assignments

Reinforce core concepts through weekly assignments. Build, solve, and assess models. Practice advanced data science techniques, and communicate results that are clear and meaningful to stakeholders.



Discussions and Interactive Activities

Participate in hands-on activities, including practicing calculations, solving problems with regression and classification, and applying techniques such as data augmentation, transfer learning, and data filtering. Get firsthand experience with Google Colab, and exchange ideas with learning peers from around the world.

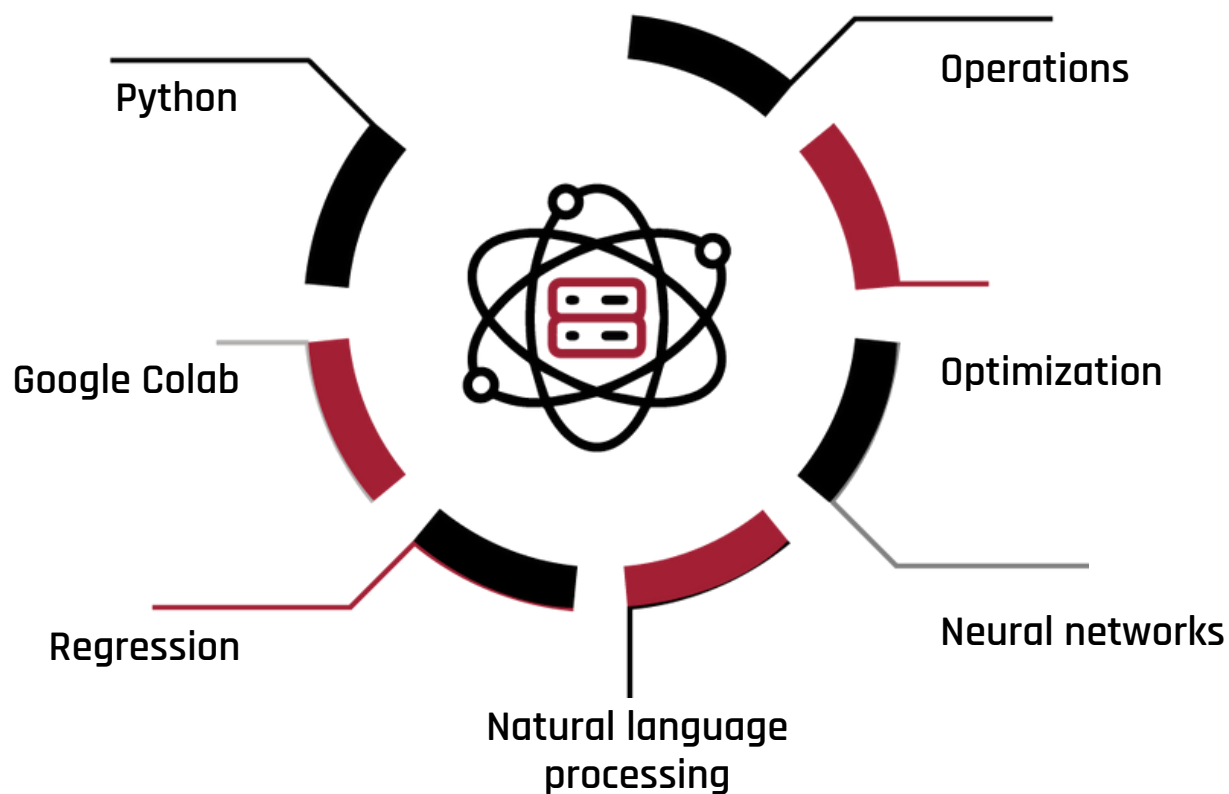
Experiential Projects

Final Capstone Project

Highlight your learnings in a final capstone project by building a portfolio of application-based assignments you have completed throughout the program. Your portfolio will demonstrate your proficiency in a range of data science skills required for model optimization and strategic decision-thinking in organizational contexts.

Program Tools and Applications

Get hands-on experience with the latest data science tools and applications, including:



Faculty



Vivek Farias

Patrick J. McGovern (1959) Professor
Professor of Operations Management
MIT Sloan School of Management

Vivek is affiliated with the Operations Management group and the Operations Research Center (ORC) at MIT Sloan School of Management. His research focuses on the development of new methodologies for large-scale dynamic optimization under uncertainty and the application of these methodologies to the design of practical revenue management

strategies across various industries, such as airlines, retail, and online advertising. He serves on the editorial boards of Management Science as a co-editor of Data Science, Operations Research, and the INFORMS Journal on Optimization.



Robert Freund

Theresa Seley Professor in Management Science
Professor of Operations Research
MIT Sloan School of Management

Robert's primary research interests are in convex optimization, computational complexity and related computational science, convex geometry, and large-scale nonlinear optimization and related mathematical systems. Robert has served as the co-editor of Mathematical Programming and as an associate editor of several

optimization and operations research journals. He received the Longuet-Higgins Prize in computer vision (2007) and numerous MIT teaching and education awards in conjunction with an MBA core course and a textbook "Data, Models, and Decisions: The Fundamentals of Management Science", co-authored with Dimitris Bertsimas. Robert received his B.A. in Mathematics from Princeton University and his M.S. and Ph.D. in Operations Research from Stanford University.



Retsef Levi

J. Spencer Standish (1945) Professor of Operations Management
Faculty Leader, MIT Leaders for Global Operations (LGO)
MIT Sloan School of Management

Retsef's research focuses on the design of analytical data-driven decision support models and tools addressing complex business and system design decisions under uncertainty in areas such as health and health care management, supply chain, procurement and inventory management, revenue management, pricing optimization, and logistics.

Retsef is the recipient of the NSF Faculty Early Career Development award, the 2008 INFORMS Optimization Prize for Young Researchers, the 2013 Daniel H. Wagner Prize, and the 2016 Harold W. Kuhn Award. Retsef received a bachelor's degree in mathematics from Tel Aviv University (Israel) in 2001 and a Ph.D. in operations research from Cornell University in 2005.



Rama Ramakrishnan

Professor of the Practice
MIT Sloan School of Management

Rama's research and teaching interests focus on the application of data science and machine learning techniques to problems in industry and the creation of intelligent products and services created by the algorithmic use of data. Rama has over two decades of experience as a data science entrepreneur and tech executive. He has founded or served

as a senior executive in four software companies that have been acquired by technology giants, such as Oracle, Salesforce, and Demandware. He remains an active member of the start-up space as an advisor, angel investor, and board member.

Rama holds a B.Tech. degree from the Indian Institute of Technology, Chennai and M.S. and Ph.D. degrees from MIT.

Career Preparation and Guidance

This program offers a wide array of career support and guidance to help you develop your career path. These services are provided by Emeritus, our learning collaborator for this program, via the Emeritus Career Center (ECC). The primary goal is to help you build the skills needed to prepare for your career, however we do not guarantee job placement. Learn more about all of the services and support available to you, including:

A SUPPORT TEAM YOU CAN RELY ON

Your support team includes program leaders and career coaches who will help you reach your learning goals and guide you through your job search.

CAREER PREPARATION SERVICES THAT GET YOU NOTICED



Write noteworthy resumes and cover letters



Create effective LinkedIn profiles



Navigate your job search



Prepare for interviews



Craft your elevator pitch



Negotiate your salary



Emeritus Career Center

Comprehensive career services. One convenient location.

The Emeritus Career Center is your one-stop shop for streamlined access to career-related services. As a participant in the program, you will gain lifetime access to the ECC and its related benefits, including:



COACHING

Schedule appointments with a career coach, and share details about new jobs and job search outcomes with them.



EVENT ACCESS

Learn about upcoming career events, register to attend, and view previously recorded webinars.



DOCUMENT STORAGE

Store your resume and other application materials in one convenient location.



RESOURCE LIBRARY

Access our growing Resource Library anytime to find job search resources, resume checklists, and other helpful information.



CAREER PROFILE

Create a profile for networking and to provide valuable information to our employer relations team and potential employers.

UNLOCK ADDITIONAL BENEFITS: SHARE YOUR RESUME

Upload your resume to the ECC for approval and take advantage of:

- A resume review and feedback from your career coach.
- Access to the job board, with postings from our employer partners and others seeking talent.
- An easy application process and the ability to track your application steps.
- Inclusion in resume books that our employer relations team will refer to employers.

Financing Options

We want to make sure that the Professional Certificate in Data Science and Analytics is an affordable option for all. This is why we offer you many different ways to pay for the program.

Loan Partners (For US Residents)

Climb Credit

Immediate repayment, interest-only repayment, and deferred payment options are available.

- Visit the [Climb Credit application portal](#)
- Fill in your basic details and proceed to the loan section of the application
- Select your course from the Program dropdown
- Choose your preferred repayment option and enter financial information
- Agree to the disclosure and submit your application
- Once approved, sign the Master Promissory Note
- Our program advisors will contact you for a confirmation on your loan application
- After confirmation, we will certify your loan. You will receive a welcome email with login instructions from notifications@instructure.com within 3 business days

Sallie Mae

Fixed repayment, interest-only repayment, and deferred payment options are available.

- Visit the [Sallie Mae application portal](#)
- Fill in your basic details and proceed to the loan application page
- At the time of loan application, please select '**Student and Career Training School**' when prompted
- Choose from fixed repayment, interest-only repayment, and deferred payment options and submit your application
- Our program advisors will contact you for a confirmation on your loan application*
- After confirmation, we will certify your loan. You will receive a welcome email with login instructions from notifications@instructure.com within 3 business days

- Visit the [Ascent Funding application portal](#)
- Enter your email address and select the 'MIT xPRO Professional Certificate in Data Science and Analytics' option from the dropdown list
- Choose from immediate repayment, interest-only repayment, and deferred payment options and submit your application
- Our program advisors will contact you with a confirmation on your loan application*
- After confirmation, we will certify your loan and you will sign the final disclosures
- You will then receive a welcome email with login instructions from notifications@instructure.com within 3 business days

Flexible Payment Options (For All Countries)

- Choose to make your payment in two, three, or six [installments](#) for higher flexibility.
- Complete your application for the [Professional Certificate in Data Science and Analytics](#) and enroll for the program.

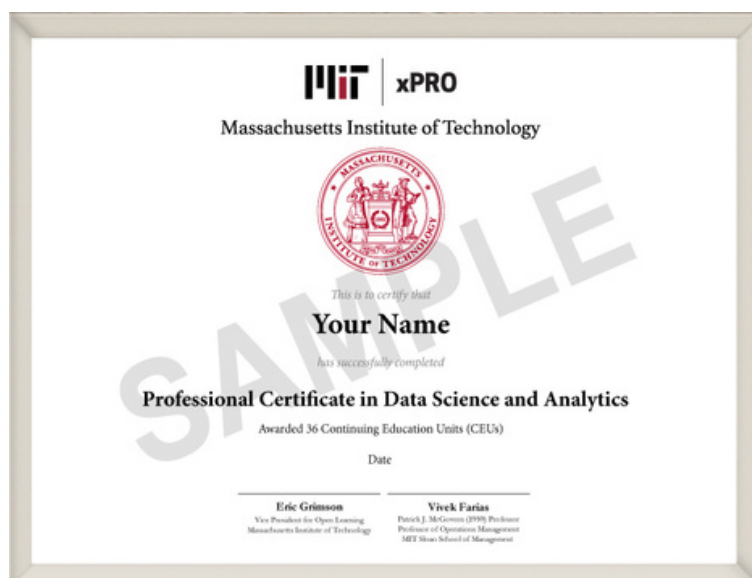
You can opt for any one of the financing options to cover up to the full cost of the program tuition. If you are considering financing your program through one of our partners, the enrollment process can only be completed with the assistance of your program advisor or by calling +1 401 443 9591.

*Due to processing time, the loan application should be submitted no later than four business days prior to the enrollment deadline.

Certificate

Get recognized. Upon successful completion of this program, MIT xPRO grants a certificate of completion to participants and 36 Continuing Education Units (CEUs). This program is graded as a pass or fail; participants must receive 75% to pass and obtain the certificate of completion.

After successful completion of the program, your verified digital certificate will be emailed to you, at no additional cost, with the name you used when registering for the program. All certificate images are for illustrative purposes only and may be subject to change at the discretion of MIT.



About MIT xPRO

MIT xPRO's online learning programs leverage vetted content from world-renowned experts to make learning accessible anytime, anywhere. Designed using cutting-edge research in the neuroscience of learning, MIT xPRO programs are application-focused, helping professionals to build their skills on the job. To explore the full catalog of MIT xPRO programs, visit xpro.mit.edu.

About Emeritus

MIT xPRO is collaborating with online education provider Emeritus to deliver this online program through a dynamic, interactive, digital learning platform. This program leverages MIT xPRO's thought leadership in engineering and management practice developed over years of research, teaching, and practice.

You can schedule a call with a program advisor from Emeritus to learn more about this MIT xPRO program.

[SCHEDULE A CALL](#)

You can apply for the program here:

[APPLY](#)



CONNECT WITH A PROGRAM ADVISOR

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UK: +44 189 236 2347

SG: +65 3129 7174

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